

AI for Healthcare

Week-1

Assignment-1

Task-1

- 1) The idea that pixels are smallest unit of digital imaging really fascinates me, the fact that images are nothing but grids of numbers is really eye-opening, earlier it was felt that image processing is quite a simple task just done by camera in our mobile phones but now I realise how complex it could be.
- 2) Secondly, I really admire the idea of how an image is zoomed in or zoomed out, before this assignment I did not know that whenever we zoom in or zoom out an entire set of new pixels are formed along with the existing ones being changed.
- 3) The biology behind the working of an eye is of course something new for me, the fact that the whole colour detection done by our eyes is mainly done by the 3 receptors namely short cones, medium cones and large cones is really amazing.
- 4) Also, this assignment cleared all the logic and funda behind the "Entire process of seeing" from light entering through the pupil to the inverted image being formed on the retina to the brain processing it.
- 5) And lastly this assignment opened the door about how computers understand and store the images.

Task-2

Types of Healthcare Reports

The major categories include:

Clinical/Patient Reports:- These include history and physical reports (created on admission, outlining the patient's medical history, presenting symptoms, and initial diagnosis), discharge summaries (detailing the patient's hospital stay, treatment received, and follow-up instructions), operative reports (documenting surgical procedures, pre-operative diagnoses, and post-operative care), and consultation reports (containing a specialist's evaluation and recommendations).

Diagnostic Reports:- Consists of the results of investigations. These include imaging reports (radiology, ultrasound, MRI), pathology reports, laboratory reports, and electrophysiological reports — all playing a fundamental role in identifying diseases and monitoring patient health.

Administrative & Aggregate Reports:- Commonly used examples include basic healthcare statistical reports, claims denial reports, and cost reports — used for improving customer service, quality of care, and operational efficiency.

Incident Reports:- These consist of unexpected events affecting staff or patients — including injuries, medication errors, equipment failure, and health hazards — transforming subjective experiences into objective data for risk management and regulatory compliance.

Medical Image Reports

Imaging reports provide detailed visual information about the inside of the body through medical images, and are crucial for diagnosing diseases, injuries, and anomalies. They are produced by radiologists after interpreting scans from modalities such as X-ray, CT, MRI, ultrasound, PET, and mammography.

1) **X-rays** :- An X-ray is a grayscale image where pixel intensity represents the attenuation of radiation by tissue. Dense structures like bone appear bright, while air-filled spaces like lungs appear dark. This makes X-rays naturally suited for image processing.

2) **MRI** :- MRI uses magnetic fields and radio waves to produce multi-contrast, multi-planar images. MRI generates **multi-channel volumetric data** — different sequences highlight different tissue properties. Images are typically in DICOM format, loaded as 3D arrays.

3) **CT Scan**:- CT (Computed Tomography) scan uses a rotating X-ray tube and detector array within a gantry to capture multiple, cross-sectional "slices" of the body. A computer processes these attenuation measurements using algorithms to create highly detailed 2D or 3D images of internal tissues, bones, and blood vessels.

4) **Ultrasound**:- Ultrasound technology uses high-frequency sound waves (2-20MHz) transmitted by piezoelectric crystals in a transducer to create real-time, non-ionizing images of internal body structures. It relies on coupling gel to eliminate air and transmit sound, creating images based on tissue-specific acoustic impedance, which determines how much sound is reflected versus transmitted.

Task-3

Dataset for various types of medical image reports.

1) X-rays:-

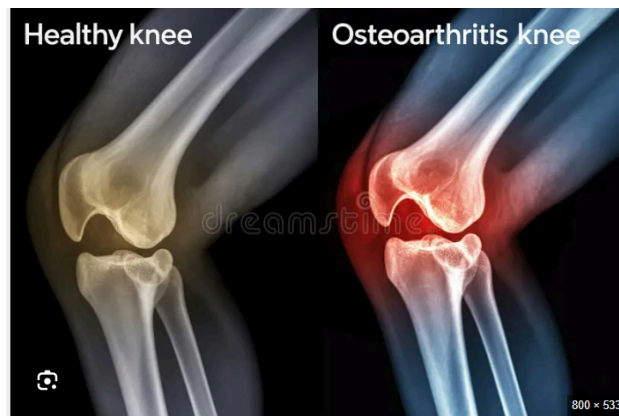
Path to import X-ray dataset from kaggle

```
import kagglehub
# Download latest version

path = kagglehub.dataset_download("paulti/chest-xray-images")

print("Path to dataset files:", path)
```

<https://www.kaggle.com/datasets/paulti/chest-xray-images>



2) MRI:-

Path to import MRI dataset from kaggle

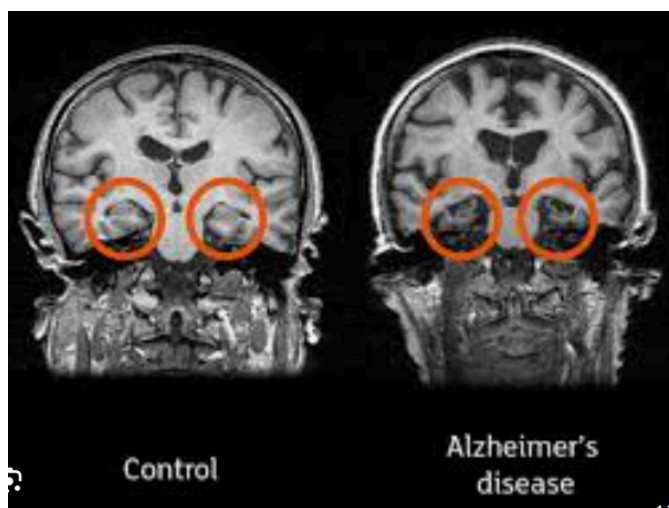
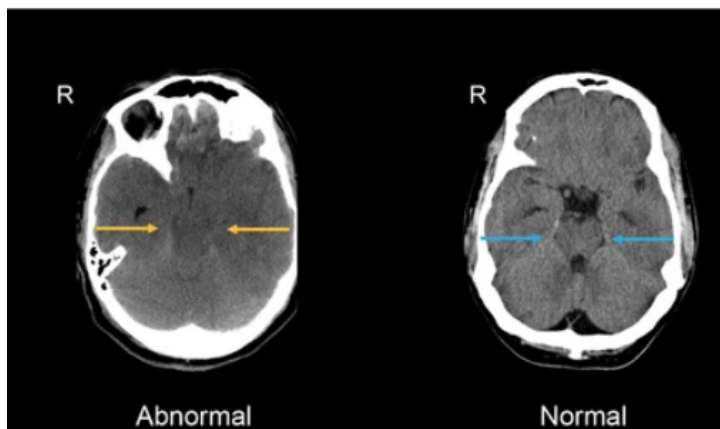
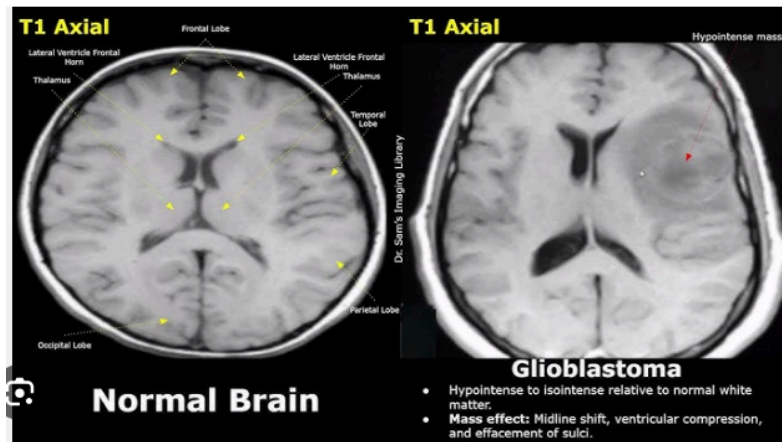
```
import kagglehub
```

```
# Download latest version
```

```
path = kagglehub.dataset_download("orville/brain-cancer-mri-dataset")
```

```
print("Path to dataset files:", path)
```

<https://www.kaggle.com/datasets/orvile/brain-cancer-mri-dataset>



3) CT Scan:-

Path to import CT Scan dataset from kaggle

```
import kagglehub

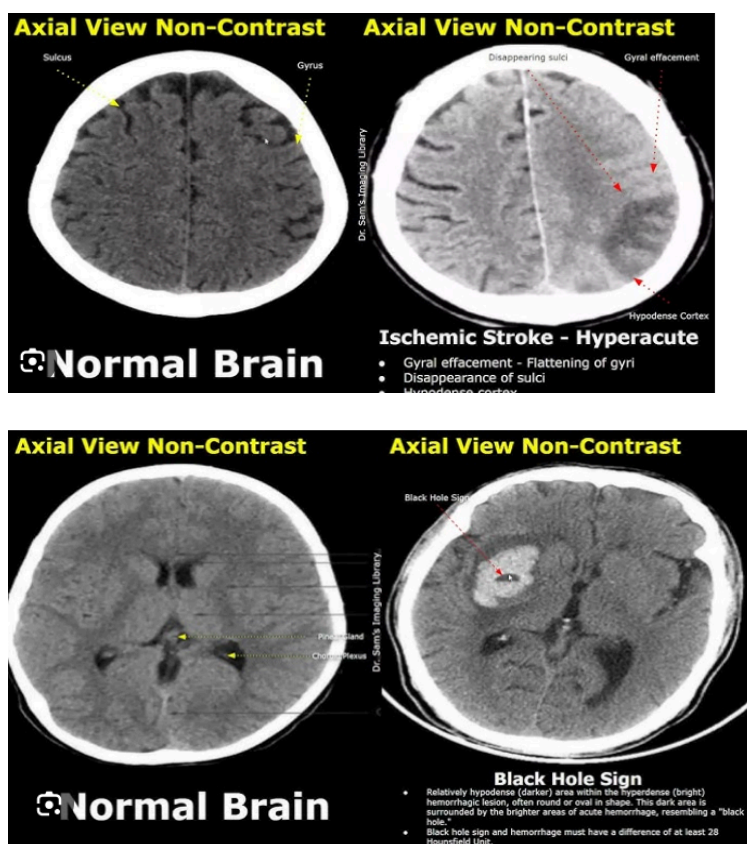
# Download latest version

path = kagglehub.dataset_download

("trainingdatapro/computed-tomography-ct-of-the-brain")

print("Path to dataset files:", path)
```

<https://www.kaggle.com/datasets/trainingdatapro/computed-tomography-ct-of-the-brain>



4) Ultrasound:-

Path to import Ultrasound dataset

```
import kagglehub

# Download latest version

path =
kagglehub.dataset_download("orville/annotated-ultrasound-liver-images-dataset")
```

```
print("Path to dataset files:", path)
```

<https://www.kaggle.com/datasets/orvile/annotated-ultrasound-liver-images-dataset>

